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CareerOrbit V3.1 — LLM-Model Registry & Prompt Engineering Bible

Audience: AI Engineers, Prompt Designers, Full-Stack Developers **Objective:** A comprehensive technical manual for the generative intelligence layer, including model selection, structured schemas, and cognitive grounding protocols.

1. Model Registry & Selection Strategy

We utilize a two-tier AI architecture to balance complex reasoning with high-speed interactivity.

T1: The Reasoning Model

(`getGPT5ReasoningModel`)

- **Model Identity:** `gpt-5.4` (State-of-the-art reasoning alias).
- **Usage:** Initial Roadmap Generation, Dynamic Recalibration, Complex Path Selection.
- **Why:** Requires deep understanding of NSQF standards, NOS codes, and BKT mastery state synchronization.

T2: The Instant Model (`getGPT5InstantModel`)

- **Model Identity:** `gpt-4o-mini`.
 - **Usage:** AI Mentor Chat, Resource Summarization, Quiz Generation.
 - **Why:** Low-latency response is critical for maintaining "flow state" during live learning.
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2. Structured Outputs & Zod Enforcement

CareerOrbit never accepts raw string responses. All AI calls are wrapped in **Zod Schemas** to ensure database compatibility.

Protocol:

- **Schema Location:** `src/lib/ai/schemas.ts`.
 - **Enforcement:** Uses the Vercel AI SDK `generateObject` and `streamObject` functions.
 - **Resilience:** If the AI fails to match the schema, the backend triggers an automatic retry (Max 2 attempts) before throwing a "Calibration Error."
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3. Cognitive Grounding: The BKT

Injection

We prevent "AI Drift" by injecting a deterministic summary of the user's knowledge state into every prompt.

The Transformation:

The `buildBKTRecalibrationSummary()` function translates the raw `knowledgeState` JSONB into a structured prompt block:

```
--- BKT KNOWLEDGE STATE ANALYSIS ---  
NEEDS FOUNDATION (pMastery < 30%):  
  ⚠ Circuit Diagnostics: 15% mastery (1 attempts)  
LEARNING ZONE (30-85%):  
  → Basic Safety: 45% mastery (2 attempts)  
--- END BKT ANALYSIS ---
```

Rule: The AI is strictly prohibited from teaching "Mastered" topics or skipping "Foundation" topics.

4. Prompt Engineering Registry

A. The Roadmap Architect

(`GPT5_ROADMAP_PROMPT`)

- **Persona:** Expert NCVET Qualification Pack (QP) Designer.
- **Constraints:**
 - Must generate 3-5 modules following the NSQF sequence (Theory -> Technical -> Readiness).
 - Must generate a verifiable `portfolio_evidence_task` for every module.
 - Must generate realistic **NOS Codes** (National Occupational Standards).

B. The Universal Guardian

(`GPT5_MENTOR_PROMPT`)

- **Persona:** Empathetic, highly encouraging, and practical AI mentor.
- **Guardian Rules:**
 - **Scope Lock:** Only discusses the roadmap and the current subtopic.
 - **Refusal:** Politely refuses topics like politics, general coding, or entertainment.
 - **Scaffolding Level:** Dynamically adjusted by the mastery engine (**maximum** to **none**).

C. The Path Selector

(**GPT5_PATH_OPTIONS_PROMPT**)

- **Persona:** Expert Career Counselor.
 - **Constraints:**
 - Must generate exactly 3 distinct paths.
 - Must factor in the user's **Device Type** and **Language Preference**.
 - Must be culturally and practically safe for the user's location and gender.
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5. YouTube Search Consistency Logic

To ensure a stable "Teaching Voice," the AI is instructed to generate queries that favor specific instructional playlists rather than random results.

- **Query Format:** [**Topic**] **Step-by-Step Full Tutorial** [**Language**]
 - **Sequential Mapping:** Uses Lesson/Part numbers for consecutive subtopics.
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8. The AI Mentor Chatbot: Technical Architecture

A. Proactive Trigger Logic (The "Attentive" Mentor)

The chatbot does not wait for the user to be confused. It monitors three behavioral signals to auto-open the UI:

1. **The 'Stuck' Signal:** Triggered if a user spends a threshold amount of time on a subtopic without progress.
2. **The 'Repeated Failure' Signal:** Triggered if the BKT engine detects multiple failed quiz attempts on the same Knowledge Component.
3. **The 'Pulsing' State:** If a trigger is active, the floating button enters an `animate-bounce` state to signal for the user's attention.

B. The Context Grounding Engine (`buildMentorContext`)

Before any message is sent to the LLM, the backend constructs a massive "Live User Context" block.

- **BKT Injection:** It pulls the exact *pMastery* and ZPD status of the *current* task.
- **Short-Term Memory:** It injects the last 5 Knowledge Components to give the AI context of the user's recent journey.
- **Scaffolding Instruction:** It converts the BKT probability into a direct command (e.g., "Give MAXIMUM scaffolding. Break into the tiniest possible steps.").

C. The Streaming Pipeline

- **Frontend:** Uses a custom `ReadableStream` reader with `TextDecoder` to provide character-by-character streaming for a "thinking" feel.
- **Voice-First Design:** Integrated with `SpeechInput` supporting multi-regional dialects (`hi-IN`, `kn-IN`, `en-IN`).
- **Stateless Persistence:** The chat history is maintained in the component state but the *Mastery Context* is re-fetched on every turn to ensure the AI always knows the user's current cognitive state.

D. Universal Guardian Rules (Scope Locking)

The LLM is locked into a **Zero-Trust Scope**:

- **Refusal Protocol:** If a user asks about non-vocational topics (politics, entertainment), the AI MUST use the standard refusal: *"As your CareerOrbit mentor, I'm here to focus on your [Path] journey..."*
 - **Safety Lock:** Strictly prohibits medical, legal, or financial advice.
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6. Language & Localization Strategy

The AI is instructed to generate **all content** in the user's `languagePreference`.

- **Prompt Instruction:** "Generate the entire response in the language specified in the user's profile context."
 - **Reliability:** High-tier models handle regional Indian languages (Hindi, Marathi, Tamil, etc.) with high fidelity when grounded in technical NOS codes.
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7. Performance & Cost Optimization

1. **Context Truncation:** We only pass the *active* roadmap module to the AI Mentor to minimize token usage.
2. **Temperature Tuning:**
 - **Roadmaps:** `temperature: 0.4` (Consistency > Creativity).
 - **Mentor:** `temperature: 0.7` (Empathy > Precision).